Task 2: Sales Data Analysis Using Hadoop MapReduce

Hadoop Streaming | HDFS | Python | Google Colab

1. Objective

In this assignment, you will gain practical experience with Hadoop MapReduce by implementing a data analysis task using Hadoop Streaming and Python.

By the end of this assignment, you should be able to:

- Explain the MapReduce programming model.
- Implement Mapper and Reducer functions.
- Execute a MapReduce job using Hadoop Streaming.
- Describe the role of HDFS in storing input and output data.
- Analyze sales data using distributed processing concepts.

2. Tools and Technologies

- Google Colab
- Hadoop
- Hadoop Distributed File System (HDFS)
- Hadoop Streaming
- Python

3. Dataset

You will work with a dataset named `sales_data.csv` containing sales transaction records.

The dataset must include the following columns:

Column	Description
transaction_id	Unique transaction identifier
region	Sales region
product_category	Product category
quantity	Number of units sold
unit_price	Price per unit

You may use the dataset provided with this assignment, or generate a sample dataset with similar columns.

4. Assignment Tasks

Task 1: Hadoop Environment Setup

1. Open the provided Google Colab notebook.
2. Install and configure the required Hadoop environment.
3. Verify the Hadoop installation.
4. Start the required Hadoop services.
5. Execute basic HDFS commands to inspect the file system.

Task 2: HDFS Operations

1. Create an HDFS directory named `/sales/input`.
2. Upload `sales_data.csv` to the HDFS input directory.
3. Verify that the file has been uploaded successfully.
4. Display the contents of the input directory.

Task 3: Implement the Mapper

Create a Python file named `mapper.py`. Your Mapper must:

1. Read the input records.
2. Calculate the revenue for each transaction using: `Revenue = Quantity × Unit Price`
3. Generate a key-value pair using: Key: `(Region, Product_Category)`, Value: `Revenue`
4. Output the generated key-value pairs in a format suitable for Hadoop Streaming.

Task 4: Implement the Reducer

Create a Python file named `reducer.py`. Your Reducer must:

1. Read the intermediate key-value pairs generated by the Mapper.
2. Group records according to their keys.
3. Calculate the total revenue for each region and product category.
4. Output the results in the following format: `Region, Product_Category, Total_Revenue`

Task 5: Execute the Hadoop MapReduce Job

1. Run your MapReduce job using Hadoop Streaming.
2. Use the HDFS input directory as the job input.
3. Store the output in an HDFS directory named `/sales/output`.
4. Verify that the job has completed successfully.
5. Display the generated output using HDFS commands.

Task 6: Analyze the Results

Based on the output your job generated, answer the following questions:

1. Which product category generated the highest total revenue in each region?
2. Which region generated the highest overall revenue?
3. How does the Mapper transform the original sales records?
4. What is the role of the Reducer in this assignment?
5. What is the purpose of HDFS in the MapReduce workflow?

5. Expected Output

Your final output should contain the total revenue for each combination of region and product category.

Example:

Region	Product Category	Total Revenue
Cairo	Electronics	15000
Cairo	Clothing	8500
Alexandria	Electronics	12000
Alexandria	Food	6000

The values above are illustrative.

7. Important Notes

- You must implement the assignment using Hadoop Streaming, not a Python-only simulation of MapReduce.
- Do not use built-in Pandas grouping operations to replace your Mapper and Reducer logic.
- Be ready to explain your implementation and the role of each MapReduce stage during the discussion.